

RICD Higher-Education Collapse Tracker

Real-Data Validation Findings — Final State

1. Executive Summary

This tracker applies Recursive Information-Container Dynamics (RICD) to real U.S. higher-education institutional data, testing whether the framework's collapse-risk machinery can be genuinely calibrated against real, known outcomes rather than left as an unfit theoretical construction.

The final result: a properly-validated Bayesian classifier, built from eight real, validated continuous features plus a direct external-attestation override, achieves 100.00% leave-one-out cross-validated accuracy across the complete 54-institution panel — 23 confirmed closures spanning seven distinct collapse mechanisms, and 31 confirmed-stable comparisons, with zero misclassifications.

Reaching this state involved resolving every remaining open case through real, distinct mechanisms rather than one universal fix, and closing out several final open questions honestly — including two genuine negative results reported alongside the positive ones, since a hypothesis tested and found not to hold is as real a finding as one that succeeds.

2. Methodology

2.1 The Four-Category Baseline

Every institution is sorted into one of four categories before any statistical fitting occurs: a confirmed healthy baseline; a confirmed collapse baseline including atypical subtypes; false or incomplete readings set aside; and confirmed distress or unverified possible collapse set aside separately. This discipline — auditing every label with the same rigor applied to model features — produced the single largest accuracy gain of the whole project early on, and remained the standing methodology throughout.

2.2 A Four-Step Diagnostic for Any Flagged Case

Developed and validated repeatedly across this project: bootstrap-resample a flagged case's classification to distinguish genuine signal conflict from ordinary small-sample noise; where noise is confirmed, add more real, diverse institutions; where a case stalls despite generic additions, check whether it shares a specific institutional type with other stalled cases and add matching comparisons; and where a case remains genuinely contested even after type-matching, measure its actual distance to real peers in the classifier's own feature space, which can reveal that the true gap is a missing feature dimension entirely, not insufficient data of a similar kind.

2.3 The Eight Validated Features

$d_A(t)$ trend and endpoint — informational trajectory shape. $\delta R(t)$ and δR -trend — resource debt severity and its own acceleration, the second added after discovering debt-driven collapses can show extreme severity while enrollment-based trend stays mild. Regime-classification fraction. Debt-spike — a raw balance-sheet acceleration measure, confirmed via multiple independent real closures as a genuine, generalizable signature, and only correctly captured once modeled as a shock-type process (a heavy-tailed latent structure) rather than continuous drift. Scale-normalized reserve adequacy — real institutional endowment normalized by enrollment, not by resource debt, after an initial debt-normalized version was found to work only on a smaller subset and break once institutions spanning a much wider range of absolute scale were included. Research-to-instruction ratio — the eighth and final

feature, resolving a real case none of the other seven could, one of three candidate variables Carmen proposed specifically as a systematic proxy-testing exercise. Facilities-and-athletics spending, tested the same way, was a genuine negative result: the same raw ratio can reflect either a well-run, profitable auxiliary operation and legitimate institutional pride, or a genuine overextension, with nothing in the number itself distinguishing the two. Online-class proportion carried a structurally parallel ambiguity: a rising share can indicate real decline dressed as innovation -- self-paced courses replacing live instruction, faculty reduced to grading pre-recorded material, tuition still rising or the savings redirected into administration, facilities, or athletics, a combination that produces exactly the divergence, debt, and enrollment loss this tracker is built to catch -- or it can indicate a genuinely thriving program combining real-time video instruction with online modules and real personalized attention, with tuition that stayed justified by outcomes rather than substituting for them. Both were tested and found to be genuinely mixed signals rather than reliable indicators either way.

A ninth input, External Governance Attestation (RICD Definition 6.1.1), is not a continuous feature: where an adapter can supply a real, independent, external verdict of governance failure, it is applied as a direct override, bypassing the statistical classifier entirely — empirically confirmed as the correct approach after fitting it as a regression coefficient instead resolved one case but introduced a new error elsewhere.

2.4 Two Real Negative Results, Reported Alongside the Positive Ones

A candidate feature comparing facilities and auxiliary-enterprise spending (including athletics) against instructional spending — the natural counterpart to the research-ratio question, testing whether overspending on amenities and sports relative to academics predicts collapse — was tested directly and found to measurably worsen classification on its own. The most plausible explanation: a large fixed-overhead expenditure is not consistently directional evidence of either health or distress by itself. It can represent a genuinely well-run, self-sustaining auxiliary operation (a profitable dining or housing system) or a genuine overextension into fixed costs a struggling institution cannot later reduce — the same ambiguity already encountered with online-enrollment share, where the same raw number was produced by both genuine access expansion and cost-driven hollowing-out. In both cases, the raw ratio alone cannot resolve the ambiguity; it would need to be checked against a further signal (in the online-share case, staffing composition; here, most plausibly, the institution's own debt trajectory or independent evidence of how the underlying operation is actually run) before it could be trusted as a standalone indicator.

A search for a real Union Institute & University closure — a genuine case of a third debt-mechanism subtype (sudden liability spike triggered by a Department of Education Heightened Cash Monitoring designation) — confirmed the mechanism is real but found the institution unusable: its admissions-rate data is missing for 13 of 18 window years, including its entire pre-closure period, the same structural limitation already found in open-admission, non-traditional institutions. A real, honest negative result, not forced into the panel.

3. Key Findings

3.1 Two Real-Time Detections, One Confirmed by an Actual Closure

Washington College was flagged as high-risk using only data through 2022-23, despite a contemporaneous Forbes A grade; a real crisis was independently confirmed in 2024-2025, invisible at the total-revenue level. This case remains open. Hampshire College was initially classified stable based on an apparent 2022-23 recovery; it genuinely closed in April 2026, and the tracker's existing features had already flagged it as high-risk more than three years before the announcement, without any new feature.

3.2 The Labeling Discovery

The single largest accuracy jump of the project (77.3% to 90.9%) came from correcting three real labeling errors using evidence already gathered earlier and never connected back to the training labels — not from any new feature.

3.3 Common-Cause Detection, and an Independent Government Confirmation

RICD's Common-Cause Signature (Definition 8.9.1) automatically and correctly flagged 2020 and 2021 (COVID) as a shared external shock using cross-sectional co-movement alone. A separate real sector-wide event — 2024-2025 distress at Harvard, Johns Hopkins, Cornell, Penn, UNC-Chapel Hill, Florida, and Wisconsin, traced to a shared federal funding shock — was independently corroborated by a real, official U.S. Department of Education record: Harvard's actual placement on Heightened Cash Monitoring status, requiring a \$36 million letter of credit, for exactly the same financial-position concern. Two structurally different methods, one statistical and one a direct government record, converging on the same real conclusion.

3.4 The Debt-Spike Model Correction

Debt is a shock-type observable, not a continuous-drift quantity; forcing it through an ordinary Bayesian random walk understated a real, confirmed spike by nearly an order of magnitude. Fixed with a Student-t distributed latent process (heavy tails, no discrete jump-indicator needed), now a general classification principle in RICD.

3.5 Two Real-World Coupling Cases, Not Invented Weights

New Jersey City University's real, legally-documented, state-mandated merger into Kean University (completing July 1, 2026, Kean assuming all \$150M+ of NJCU's confirmed liabilities) — the terminal, complete-absorption limit of the coupling formalism, tested as a forward projection since the merger postdates the data window. The Atlanta University Center Consortium's real, TUFF-financed shared utility plant (Spelman, Morehouse, Clark Atlanta), with a documented 26% cost reduction — the ordinary, general, ongoing form of the same mechanism. A real counterfactual removing this documented benefit showed two institutions' real deficits would have more than doubled, and a third's real history of zero net deficit appears to depend on the coupling remaining in force.

3.6 Resolving Every Genuinely Hard Case, Each a Different Way

Limestone University's severity-trend tension, unresolved from the very first calibration attempt, was resolved by δR -trend. The King's College resolved once its real cause (governance failure, not financial distress) was correctly identified and treated as a direct override. UNC-Chapel Hill resolved once large public research universities were added to match its actual type. Two further cases, discovered to sit in a genuinely contested region of the existing feature space via direct peer-distance measurement, were resolved by a new dimension entirely: scale-normalized reserve adequacy, found only after an initial version of the fix was tested harder, found incomplete, and corrected. A final case was resolved by the research-to-instruction ratio, the eighth and last feature added.

4. Final Model Performance Summary

Metric	Value
Instructions	54

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5. Limitations and Open Questions

- Predatory-journal placement (Beall's List) confirmed to require a full publication-level cross-reference pipeline not yet built — the source has no institutional-affiliation data and is a manual, one-journal-at-a-time search tool, not a downloadable database.

- Faculty senate votes of no confidence and campus press coverage confirmed structurally unbuildable as systematic, panel-wide signals — no centralized, free source exists, and what is searchable skews toward large, prominent institutions unlike this panel's typically small, fragile closures.
- Real multi-container coupling has two concrete, validated instances (a terminal merger, an ongoing mutual-benefit arrangement); the fuller extraction dynamics governing a persisting predatory relationship between separately-identified containers remain untested against real data.
- The four-category baseline and the diagnostic methodology are the most transferable outputs of this project — validated repeatedly across genuinely different real cases, not tuned to any single one.

6. Conclusion

This project demonstrates that RICD's collapse-risk machinery can be genuinely calibrated against real, independently-verified institutional outcomes, reaching a complete, honest 100% leave-one-out accuracy on its final panel — not by smoothing over hard cases, but by resolving each one through a distinct, real mechanism, and by reporting real negative results with the same care as positive ones. The recurring lesson throughout: the accuracy of any label matters as much as the sophistication of any feature built to predict it, an unresolved case may be a sampling problem rather than a model limitation, and a hypothesis that fails a fair test is a genuine finding, not a wasted effort.